Assignment # 3

STAT 679: Statistical Methods for Spatial Lattice Data–Spring 2026, UW-Madison

1. **(5 points)** Refer to Figure: (*"Poverty by Socio-Economic Factors in Wisconsin/Illinois in 2000"*) from the *Poverty in Wisconsin and Illinois Data Example: Exploratory Data Analysis* material. Interpret what you observe in the visualization for the other seven socio-economic factors.

2. **(10 points)** Refer to the *Poverty in Wisconsin and Illinois Data Example: Simple Linear Regression* material.

   (a) Complete the interpretation for all other regression coefficients for `m1`.

   (b) Complete the interpretation for all other regression coefficients for `m1.back`.

   (c) Perform stepwise regression using `direction = "both"`. How do the results change, if at all?

3. Consider the following grid of experimental plots:

1	2	3	4	5
6	7	8	9	10
11	12	13	14	15
16	17	18	19	20
21	22	23	24	25
26	27	28	29	30

   The data in `corn.csv` consist of organic phosphorus in corn (y), organic phosphorus in soil (x_1), and inorganic phosphorus in soil (x_2) for each of the plots in the grid. Consider the following models:

$$\text{Model I: } y = \beta_0 + \beta_1 x_1 + \varepsilon$$

$$\text{Model II: } y = \beta_0 + \beta_2 x_2 + \varepsilon$$

$$\text{Model III: } y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$$

   (15 points) For each model:

   (a) Perform exploratory spatial data analysis on the data.

   (b) Fit the model using the usual regression approach, ignoring spatial considerations. Construct and interpret a residual plot.

   (c) Perform and interpret a test of significance for the slope(s) and intercept in the regression model.

4. **(5 points)** For the three models in Question 3, test spatial correlation among the residuals, assuming a rook's case neighborhood and binary spatial weights, using `moran.test()`. Do this for all three models.

   Hint: The following R commands may be helpful:

```
corn.lm <- lm(y ~ x1, data = corn)
summary(corn.lm)
plot(corn.lm$fitted, corn.lm$resid)
moran.test(corn.lm$resid, mylw, alternative = "two.sided")
```

The `lm` function fits a linear model. In the case above, a linear model of y versus x_1 is fit. The `summary()` and `anova()` functions produce useful regression output. Note how the residuals and predicted values are accessed using `corn.lm$resid` and `corn.lm$fitted`.

5. **(5 points)** A supposedly better test, although the mathematics is more complex, is implemented in `spdep` as follows:

   ```
   lm.morantest(corn.lm, mylw, alternative = "two.sided")
   ```

   Perform this test for the three models in Question 3, and compare the results with those from Question 4. Do this for all three models.

6. **(5 points)** Interpret your findings for the problem as a whole using the likelihood ratio test. For the three models in Question 3, test either x_1 or x_2 by comparing a reduced model to the full model.

   Hint: The following R command may be helpful:

   ```
   anova(corn.lm_reduced, corn.lm_full, test = "Chisq")
   ```